

Kevin Rychel-Penn, PhD

Computational Biology | Spatial & Multi-Omics | Biological Data Science

San Diego, CA | kevinrychel.com | kevinrychelpenn@gmail.com | linkedin.com/in/kevin-rychel

PROFILE

Computational biologist and data scientist specializing in spatial biology, multi-omics, quantitative imaging, and scientific software. Builds reproducible analysis systems that turn complex biological measurements into interpretable insight, spanning experimental design, QC, statistical modeling, machine learning, software deployment, and scientific communication. Combines deep technical work with a record of making complex methods clear and actionable for computational, experimental, customer, and leadership audiences.

TECHNICAL EXPERTISE

Biological data & methods: spatial proteomics, multiplex immunofluorescence, snRNA-seq/transcriptomics, multi-omic integration, cell typing, clustering/dimensionality reduction, spatial neighborhoods, differential/compositional statistics, segmentation, registration, batch/QC analysis

Scientific computing: Python; Pandas, NumPy, SciPy, scikit-learn, Matplotlib, Seaborn, Plotly, Scanpy, Squidpy, Napari, Cellpose, StarDist, MapMyCells, PyTorch (working exposure); R (working exposure: spicyR, sccomp)

Workflow engineering & infrastructure: Git, Jupyter, Linux/WSL, Bash, Nextflow, Snakemake, Docker, AWS/S3 (working exposure); AI-assisted software development with Codex and Claude Code

Scientific platforms: MACS iQ View, Imaris

PROFESSIONAL EXPERIENCE

Miltenyi Biotec — Data Scientist, Spatial Biology | San Diego, CA

Nov 2023 – Present

- Serve as the sole data scientist within the US imaging applications-development team, owning advanced analysis for customer feasibilities, academic collaborations, internal R&D, and spatial-biology application development.
- Build scalable QC, cell-typing, and statistical workflows for TB-scale multiplex studies with hundreds of thousands of cells and ~20–80 protein markers, accounting for segmentation, intensity, batch, and ROI effects.
- Architected and implemented DAWN/DUSK, a Python ecosystem for reproducible human-in-the-loop batch processing, cohort-level statistics, and visualization; distributed to 50+ customer users, with planned MACS iQ View integration.
- Lead computational analyses across immunology, oncology, and neurodegeneration; in glioblastoma, cell-state and neighborhood modeling identified a candidate mutation-dependent stem-like proliferative niche.
- Optimized registration workflows and hyperparameters for 3D light-sheet mouse-brain imaging to support robust Allen Brain Atlas annotation; built novel Napari visualizations and advised R&D on product implementation.
- Develop multi-omic methods spanning RNA/protein imaging and snRNA-seq, including a machine-learning panel-design workflow to select informative targets for multiplex spatial imaging and cross-modality cell typing.
- Delivered custom analysis that materially supported a ~\$200K pharmaceutical instrument purchase; the customer continues to use the analysis software as an important extension of the platform.
- Launching an internal DAWN subject-matter-expert group; define advanced-analysis standards, training materials, and support for clustering, dimensionality reduction, Python workflows, and customer troubleshooting.
- Teach spatial data science at US MACSima customer trainings and through hands-on user-group workshops, webinars, and industry events; a recent webinar drew 170+ registrations and 60+ attendees.

Palmona Pathogenomics — Genomics Data Scientist | San Diego, CA

May 2023 – Nov 2023

- Designed and implemented a machine-learning workflow for rapid development and deployment of models predicting antimicrobial-resistance phenotypes from genomic datasets.
- Built reproducible Nextflow/Docker workflows and contributed to bioinformatics web-app design; containerized code was later deployed on a private global-health data-processing platform.

UC San Diego, Systems Biology Research Group — PhD Researcher & Genome Analytics Group Leader

Sep 2019 – Jun 2023

- Applied machine learning and large-scale transcriptomics to infer 12+ transcriptional regulatory networks from public RNA-seq datasets; built reusable pipelines and analysis packages for collaborative systems-biology research.
- Co-led the ~17-member Genome Analytics subgroup, recruiting 10 graduate researchers, mentoring trainees, coordinating international collaborations, and representing the team in funding reports and scientific presentations.
- Developed iModulonDB (>300 monthly users), leading to first-author papers in Nature Communications and Nucleic Acids Research.
- Received the 2023 Engelson PhD Thesis Award for best overall UC San Diego Bioengineering dissertation.

SELECTED SCIENTIFIC IMPACT

- 30+ peer-reviewed publications • 4 first-author papers • 1,000+ citations • h-index 20
- Rychel K, Sastry AV, Palsson BO. Machine learning uncovers independently regulated modules in the *Bacillus subtilis* transcriptome. *Nature Communications* (2020).
 - Rychel K, et al. iModulonDB: a knowledgebase of microbial transcriptional regulation derived from machine learning. *Nucleic Acids Research* (2021).
 - Rychel K, et al. Lab evolution, transcriptomics, and modeling reveal mechanisms of paraquat tolerance. *Cell Reports* (2023).
 - Rychel K, et al. Laboratory evolution reveals transcriptional mechanisms underlying thermal adaptation of *Escherichia coli*. *Genome Biology and Evolution* (2025).

SELECTED PRESENTATIONS & SCIENTIFIC COMMUNICATION

- Society for Neuroscience Annual Meeting (2026) — accepted poster integrating snRNA-seq, 2D spatial proteomics, and 3D light-sheet imaging.
- Bitesize Bio webinar (2026) — scaling spatial-biology analysis from many markers to many datasets; 170+ registrations and 60+ attendees.
- SelectScience webinar (2025) — interpreting complex spatial biology data through unsupervised data analytics; 160+ registrations and 40+ attendees.
- MACSima customer education (2024–2026) — recurring US trainings, hands-on user-group workshops, and invited sessions across New York, San Francisco, San Diego, Montana, Germany, the UK, and France.
- Graduate research — two podium talks representing the Genome Analytics subgroup at DTU’s Center for Biosustainability (Copenhagen) and a poster at Gordon Research Conference on Microbial Stress Response.

EDUCATION

University of California San Diego — PhD, Bioengineering GPA 3.97/4.00	2018 – 2023
University of Michigan — BSE, Biomedical Engineering; Minor, Computer Science GPA 3.98/4.00	2014 – 2018

HONORS & INTERESTS

Honors: Engelson PhD Thesis Award • Summa Cum Laude • Tau Beta Pi • James B. Angell Scholar • National MS Society Scholarship • UC Grad Slam Finalist

Interests: Yoga, running, weight-lifting • International travel • Mixology • Astronomy/cosmology • Intermediate Spanish